

PART (A) Breast Cancer Mortality Prediction

Task (1) – Domain Understanding: Classification.

Variable Name	Retain or Drop	Brief justification for retention or dropping
Patient ID	Drop	Has no effect on the target variable (cancer mortality).
Month of Birth	Drop	Has no effect on target variable(cancer mortality).
Age	Retain	Age influences cancer mortality (American Cancer Society. Cancer Facts & Figures 2024 Atlanta: American Cancer Society; 2024).
Sex	Drop	All the patients are female.
Race	Retain	Mortality rates differ across the races(Siegel RL, Miller KD, Jemal A. Cancer statistics, 2020. CA Cancer J Clin. 2020;70(1):7-30.).
Marital Status	Retain	Research shows that married people have a lower cancer mortality rate. (Chen, X., Chen, Y., & Yang, Y. (2023). Marital status and cancer survival: A meta-analysis of prospective cohort studies. Annals of Epidemiology, 78, 11-18.)
Occupation Code	Retain	Particular job types expose you to higher cancer risk. (International Agency for Research on Cancer. IARC Monographs on the Evaluation of Carcinogenic Risks to Humans: Asbestos. Lyon, France: IARC; 1987. Volume 43.)
Adopted Status	Retain	Adoptees may have increased risk of cancer(Pafitis, N. (2018). Are mortality rates affected by age adoptees t to their adoptive families? BMC Public Health, 18(1), 1-8
T Stage	Retain	A higher value increases cancer mortality.
N Stage	Retain	A higher value increases cancer mortality.
6th Stage	Retain	A higher value increases cancer mortality.
Differentiate	Retain	The growth and look of the cancer cells have an effect on cancer mortality.

Grade	Retain	A higher value increases cancer mortality.
A Stage	Retain	A higher value increases cancer mortality.
Tumour Size	Retain	A higher value increases cancer mortality.
Estrogen Status	Retain	Positive estrogen status can be treated with hormone therapy thereby reducing cancer mortality.
Progesterone Status	Retain	Positive progesterone status can be treated with hormone therapy thereby reducing cancer mortality.
Regional Node Examined	Retain	Determines if the cancer has spread to lymph nodes.
Regional Node Positive	Retain	Confirms if cancer has spread to lymph nodes.
Survival Months	Retain	Target variable
Mortality Status	Retain	Target variable

Task (2) – Data Understanding: Producing Your Experimental Designing

```
[ ] df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4024 entries, 0 to 4023
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Patient ID            4024 non-null   object
 1   Month of Birth        4024 non-null   int64
 2   Age                   4015 non-null   float64
 3   Sex                   3947 non-null   object
 4   Race                  4024 non-null   object
 5   Marital Status        4024 non-null   object
 6   Occupation Code       4021 non-null   float64
 7   Adopted Status        4024 non-null   int64
 8   T Stage               4024 non-null   object
 9   N Stage               4024 non-null   object
10   6th Stage             4024 non-null   object
11   differentiated         4024 non-null   object
12   Grade                 4024 non-null   int64
13   A Stage               4024 non-null   object
14   Tumor Size            4021 non-null   float64
15   Estrogen Status       4024 non-null   object
16   Progesterone Status   4024 non-null   object
17   Regional Node Examined 4023 non-null   object
18   Regional Node Positive 4024 non-null   object
19   Survival Months       4024 non-null   int64
20   Mortality_Status      4020 non-null   object
dtypes: float64(3), int64(4), object(14)
memory usage: 660.3+ KB
```

df.describe()

	Moth of Birth	Age	Occupation Code	Adopted Status	Grade	Tumor Size	Survival Months
count	4024.000000	4015.000000	4021.000000	4024.000000	4024.000000	4021.000000	4024.000000
mean	6.481362	54.094396	8.427257	0.496769	2.150596	30.419299	71.472167
std	3.475442	11.417118	4.637345	0.500052	0.638234	21.161080	25.361855
min	1.000000	30.000000	1.000000	0.000000	1.000000	-75.000000	1.000000
25%	3.000000	47.000000	4.000000	0.000000	2.000000	16.000000	56.000000
50%	6.000000	54.000000	8.000000	0.000000	2.000000	25.000000	73.000000
75%	10.000000	61.000000	12.000000	1.000000	3.000000	38.000000	90.000000
max	12.000000	502.000000	16.000000	1.000000	4.000000	140.000000	760.000000

a) Investigate any issues in your retained dataset and the possible variables.

Variable Name	Issue description	Proposed mitigation	Justification for used mitigation
Age	Missing Values	Remove missing values using dropna method.	Can introduce bias into the model, removing it improves accuracy.
Sex	Missing Values	Remove missing values using dropna method.	Can introduce bias into the model, removing it improves accuracy.
Tumor Size	Missing Values	Remove missing values using dropna method.	Can introduce bias into the model, removing it improves accuracy.
Regional Node Examined	Missing Values	Remove missing values using dropna method.	Can introduce bias into the model, removing it improves accuracy.
Mortality_Status	Missing Values	Remove missing values using dropna method.	Can introduce bias into the model, removing it improves accuracy.
Occupation Code.	Missing Values	Remove missing values using dropna method.	Can introduce bias into the model, removing it improves accuracy.
Mortality_Status	Casing inconsistency.	Change all the values to lower case using lower method.	Ensures data is interpreted correctly.
Age	Age higher than 100	Remove the age using the drop method.	Can skew the data, therefore introducing bias.
Moth of Birth	Spelling error	Rename it to MOB using rename function.	Enhances accurate model building processed.
T stage	White space	Remove white space using the rename method.	Enhances accurate model building processed.
Reginol Node Positive'	Spelling error	Rename it using rename method.	Enhances accurate model building processed.

b) With the aid of Python packages and a notebook, implement your suggested mitigations of issues in Task (3.a), and show evidence of implementing your suggested solutions to the problems you identified for your dataset in (Task 3. a).

```
[ ] df_clean['Mortality_Status'] = df_clean['Mortality_Status'].str.lower()

[6] df_clean = df.dropna()

[7] df_clean = df_clean.drop(522)

df_clean.shape
(3927, 21)

[8] df_clean = df_clean.rename({'Patient ID ':'ID', 'Moth of Birth':"MOB", 'T Stage ':'T Stage',
                               'Reginol Node Positive': "Regional Node Positive",}, axis=1)

[ ] df_clean.columns
Index(['ID', 'MOB', 'Age', 'Sex', 'Race', 'Marital Status', 'Occupation Code',
       'Adopted Status', 'T Stage', 'N Stage', '6th Stage', 'differentiate',
       'Grade', 'A Stage', 'Tumor Size', 'Estrogen Status',
       'Progesterone Status', 'Regional Node Examined',
       'Regional Node Positive', 'Survival Months', 'Mortality_Status'],
      dtype='object')
```

Task (4) – Modelling: Create Predictive Classification Models

a) List down the type of each algorithm (parametric vs non-parametric), name any learnable parameters, and list any possible hyperparameters for each algorithm which you may want to consider tuning.

Algorithm Name	Algorithm Type	Learnable Parameters	Some Possible Hyperparameters	Imported Python package to use the algorithm
NB	Parametric	- Class priors - Conditional probabilities	- K - Distance metric	from sklearn.naive_bayes import GaussianNB
LR	Parametric	- Weights (coefficients) - Bias (intercept)	Laplace smoothing	from sklearn.linear_model import LogisticRegression

KNN (N=?)	Non-parametric	-	- Regularization strength (C) - Optimizer Learning rate	<pre>from sklearn.neighbors import KNeighborsClassifier</pre>
--------------	----------------	---	--	---

b) With the aid of the Python packages, use the training–test split approach with your retained applicable categorical input features only and the class output feature to build your predictive classification models.

i. Screenshot the list of all feature names used for building your classification models and the corresponding data shape function output.

[49] `encoded_df.columns`

```
Index(['num__Age', 'num__Occupation Code', 'num__Adopted Status', 'num__Grade',
      'num__Tumor Size', 'num__Survival Months', 'cat__Race_Black',
      'cat__Race_Other', 'cat__Race_White', 'cat__Marital Status_Divorced',
      ...,
      'cat__Regional Node Positive_4', 'cat__Regional Node Positive_41',
      'cat__Regional Node Positive_46', 'cat__Regional Node Positive_5',
      'cat__Regional Node Positive_6', 'cat__Regional Node Positive_7',
      'cat__Regional Node Positive_8', 'cat__Regional Node Positive_9',
      'cat__Regional Node Positive_One', 'Mortality_Status'],
      dtype='object', length=132)
```

[50] `encoded_df.shape`

```
(3927, 132)
```

ii. In less than 100 words, research and justify your choice of the training-test split ratio and provide an in-text citation.

A 70-30 train-test split provides enough data for the model to learn effectively and generalize well.

Kuhn, M., & Johnson, K. (2013). Applied predictive modeling. Springer.

iii. In less than 100 words, discuss the overall purpose of using a training-test approach in contrast to the use of validation sets in K-fold cross-validation and describe the case/s when to apply each of those approach is used.

Train-test split is used for quick model evaluation with large datasets, dividing data into training and testing sets while K-fold cross-validation, suitable for limited data, uses multiple splits for robust performance estimation and model selection.

iv. Provide as evidence the code line from your source code that ensures that all models were tested on the same test dataset, also ensure that the labels ratio of Mortality Status “Alive” to Mortality Status “Dead” is the same in the training and test sets.

```

from sklearn.model_selection import train_test_split

X1_train,X1_test,y_train,y_test=train_test_split(X1,y,test_size=0.30,random_state=44, stratify = y)

```

Task (5) – Evaluation: How good are your models.

- a) With the aid of Python packages, paste the test confusion matrix for each trained model as screenshots from the output of your Python code.
- b) Five different classification evaluation metrics are noted. Paste each model's test performance results.

Metrics	USE or DO NOT USE	Justification in relation to the success criteria	Model Name	Test Score
Accuracy	DO NOT USE.	The success criteria requires the model to predict a higher portion of actual positive cases and this metric is not effective in measuring that.	NB	0.4639
			LR	0.8868
			KNN (K=?)	0.8836
Recall	USE	The success criteria requires the model to predict a higher portion of actual positive cases and this metric is effective in measuring that.	NB	0.4639
			LR	0.8868
			KNN (K=?)	0.8836
Precision	DO NOT USE.	The success criteria requires the model to predict a higher portion of actual positive cases and this metric is not effective in measuring that.	NB	0.8201
			LR	0.8738
			KNN (K=?)	0.8701
F-Score	DO NOT USE.	The success criteria requires the model to predict a higher portion of actual positive cases and this metric is not effective in measuring that.	NB	0.5841
			LR	0.8726
			KNN (K=?)	0.8697
AUC-ROC	DO NOT USE.	The success criteria requires the model to predict a higher portion of actual positive	NB	0.00
			LR	0.00

		cases and this metric is not effective in measuring that.	KNN (K=?)	0.00
--	--	---	-----------	------

c) Suggest a single best classification model based on the 'USED' performance metrics scores you identified in (Task 5. b). Briefly describe how well your best model satisfies the needs of your healthcare professionals.

d) Investigated with evidence to establish whether your selected best model is good fit, underfit or overfit.

e) To enhance your selected best model/s performance, tune some of its possible hyperparameters, which you indicated in (Task 4. a) for that specific algorithm. With the aid of Python packages, Re-train the algorithm again with GridSearchCV

i. Indicate the number of cross-validation K folds used.

No. of k folds used: 5

ii. For the newly tuned model, document the estimated best hyperparameters,

iii. Present the test confusion matrix for the best models before and after tuning.

iv. Calculate and document the new score/s of the "USED" performance metric/s of your choice to interpret the success criteria identified in (Task 5.b) before and after tuning.

v. Use your observations to comment on whether the tuning of hyperparameters of your best model improved its positive predictive ability in line with the success criteria.

f) Based on your best model, draft an answer for the research question, criticise your best-performing model, and state any limitations you may have identified. Research and try to explain why your selected algorithm overtook all other models in no more than 100 words. State any ethical issues your model may raise if used to screen for breast cancer mortality.

PART (B) Breast Cancer Survival Rate Prediction

Task (1) – Domain Understanding: Regression.

Using python functions, show the dimensions of your data subset that you will RETAIN for this regression modelling problem. Using python functions, list the names of the features that you intend to use for modelling from Table.1.

```
[53] encoded_df.shape
```

```
(3927, 132)
```

```
feature_cols_DT = encoded_df.drop('num__Survival Months', axis = 1).columns  
feature_cols_DT
```

```
Index(['num__Age', 'num__Occupation Code', 'num__Adopted Status', 'num__Grade',  
      'num__Tumor Size', 'cat__Race_Black', 'cat__Race_Other',  
      'cat__Race_White', 'cat__Marital Status_Divorced',  
      'cat__Marital Status_Married',  
      ...,  
      'cat__Regional Node Positive_4', 'cat__Regional Node Positive_41',  
      'cat__Regional Node Positive_46', 'cat__Regional Node Positive_5',  
      'cat__Regional Node Positive_6', 'cat__Regional Node Positive_7',  
      'cat__Regional Node Positive_8', 'cat__Regional Node Positive_9',  
      'cat__Regional Node Positive_One', 'Mortality_Status'],  
      dtype='object', length=131)
```

```
[32] X = encoded_df[feature_cols]
```

```
y = encoded_df['num__Survival Months']
```

Task (2) – Data Understanding: Producing Your Experimental Designing

Plot the distribution for your RETAINED input variables and your “target” variable.

Task (3) – Data Preprocessing: Transforming your data

a) By looking at the dataset establish whether there is a need for scaling your dataset attributes. Explain with evidence from your python code output the reasoning behind your recommendation.

b) In general, when applying scaling to any dataset for regression modelling, would scale the input features only, the target feature only, or all features? In less than 150 words, briefly justify your answer and include in-text citation where appropriate.

Task (4) – Modelling: Build Predictive Regression Models

a) In less than 50 words, explain the added benefit of using a DT regressor to this healthcare prediction problem.

b) With the aid of the Python packages, you will use training – test split of 80:20 to build and test two DT regression models, Model 1 & Model 2. The first DT model with numeric features only and the second model using all your retained features:

i. From your python notebook, insert in your report, provide as evidence the code line from your source code that ensures reproducibility of your training - test sampling.

ii. Using python packages, show from your code output the dimensions of your training and test subsets used for each model. List the subset of features names used for Model 1 and Model 2.